Supplementary Material

Machine learning in battery management systems: a two-tier scoping review of 799 eligible studies with in-depth classification and validation audit

This supplement consolidates the robustness material that supports the main text: the interpretation rules that govern every quantitative claim (S1), the label-noise scenario envelope for the validation counts (S2), the minimum-detectable-effect analysis of the method-family contrast (S3), the construction and limits of the recall audit (S4), the two abstract-level probes (S5), the descriptive conference-venue probe (S6), and the per-paper evidence of the validation audit (S7). Every section states the released artifacts it derives from; the complete evidence base and analysis pipeline are deposited at Zenodo, DOI 10.5281/zenodo.23436202, with the GitHub repository of the same name as a development mirror. The main text carries one-sentence summaries of each item with pointers here, so the supplement can be read alongside Sections 2.2–2.3, 2.5–2.6, and 3.4 of the extended review (Sections II-A–II-C and III-A–III-C of the condensed submission manuscript).

S1. Interpretation rules for every quantitative claim in the paper

Five rules govern how the paper's numbers may be read. They are stated here in full because the main text invokes them by number.

Rule 1 (subset-scoped claims). Anything coded from full texts — function, method family, chemistry, data source, validation protocol, baseline screen — describes the 103-study classified subset only and is never generalized to the field. Headlines that read 'the subset's' or 'within the classified subset' invoke this rule.

Rule 2 (pool-scoped claims). Tier-1 counts (799 records) measure a capped 30-query Crossref retrieval, not a census. Per-year counts are observed counts inside that cap; query-theme tag shares are read for ordering, not exact rates; the 2024 window is partially covered (no query pass was dedicated to January–May 2024); and the recall audit of S4 (16–17% capture of the reviews' own cited literature) bounds external validity.

Rule 3 (earliest means earliest in the classified subset). Recall is bounded, so 'first' and 'earliest' claims are subset-first by construction and are phrased accordingly.

Rule 4 (non-significance is not equivalence). The physics-informed versus data-driven contrast is underpowered (minimum detectable effect roughly 30–35 percentage points, S3); 'no detectable difference' is reported as exactly that.

Rule 5 (curation conditions the subset). Tier-2 selection is quota- and citation-ranking-driven (main text, Section 2.3), so distributions inside the 103 partly reflect those choices and are reported as corpus composition.

Invoked at: extended-review Sections 2.2, 2.3, 2.6, 3.2–3.4, and the Figure 5 caption (condensed-manuscript Sections II-A, II-C, and III-A–C). Released artifacts: selection.json, screening_log.csv, pool.csv, recall_audit.json.

S2. Label-noise scenario envelope for the validation counts

The strong/weak validation counts of the main text rest on a single full-text coding by the lead author (the two-round reliability study of Section 2.5 measures abstract-level reproducibility in round 1 and rule-application agreement on a shared evidence base in round 2 — neither is an independent full-text dual coding of the labels). To quantify how much those counts depend on the labels, a label-noise scenario analysis perturbs them directly. The analysis is a scenario envelope, not an error model.

Design. In each of 4,000 seeded Monte Carlo draws (seed 7), a label-flip rate is drawn from a Normal distribution centered on an observed coder-versus-label error rate with standard deviation 0.02, truncated at zero; the drawn rate then flips strong/non-strong bucket labels across the 88 primary papers, and both the overall strong share and the physics-versus-other Fisher exact test are recomputed.

Scenario 1 (symmetric). The rate is centered on the coders' overall bucket error rate against the released labels (0.37 and 0.27 for the two coders; mean 0.32), and flips labels in either direction. Symmetric random flipping mechanically attenuates contrasts toward the null.

Scenario 2 (strong-to-weak, worst-case bound). The actually observed coder-label disagreements are directional and concentrated on the strong labels: 16 of the 19 bucket disagreements have the coder assigning weak where the released label says strong (the declared conservative fallback of Section 2.5), 13 of the 30 sampled papers are strong-labelled, and each coder recoded 8 of those 13 weak — a strong-label-specific rate of 0.62 (16 of 26 pooled coder-labels). Scenario 2 therefore flips strong labels to weak only, at a rate drawn from a Normal centered on that strong-label-specific rate. Because the coders worked from abstracts under a declared weak-if-unreported rule, that rate is an upper bound on plausible label error, and Scenario 2 is reported as a worst-case bound, not an estimate.

Results. Under the symmetric scenario the strong-protocol count spans 35–52 of the 88 primary papers (39.8–59.1%) and the family contrast remains non-significant in 94.4% of draws. Under the strong-to-weak worst-case bound the count falls to about 16 of 88 (≈18%) at the observed recode rate, with a scenario envelope of 10–23 (11.4–26.1%), and the contrast remains non-significant in 97.6% of draws. Read together, the two scenarios bracket the single-coder counts around the documented disagreement structure — under either, the physics-versus-other contrast shows no detectable difference and remains underpowered in the sense quantified in S3 — and the worst-case bound says only how low the strong-protocol count could sit if abstract-level recoding were taken at face value, not where it sits.

Why round 2 does not move this envelope. The round-2 re-coding under the revised codebook (Section 2.5) lifts between-coder agreement on the strong/non-strong bucket to 28 of 30 ($\kappa = 0.83$), but its coder-versus-label comparison is mechanically capped — the v1.1-only codes (not-ML, production-and-quality, field-stationary, not-reported) cannot match the v1.0 released-label vocabulary by construction, and broader chemistry classes cannot match released specifics — so it yields no clean recode rate to replace the round-1 rates above. The envelope therefore remains built on the round-1 independent-evidence disagreement structure; what round 2 bounds is rule ambiguity (which the v1.1 repair largely removed), not label error.

Released artifacts: sensitivity_label_noise.py and sensitivity_label_noise.json; gold_labels_30.csv (the 30-sample labels); coder_B_30.csv, coder_C_30.csv, agreement_report.csv, dispute_items.csv, and the hash-tracked reliability log under corpus/reliability/; round 2: coder_B_30_fulltext.csv, coder_C_30_fulltext.csv, agreement_report_fulltext.csv, dispute_items_fulltext.csv.

S3. Minimum-detectable-effect analysis of the method-family contrast

The main-text contrast — 8 of 22 physics-informed/hybrid primary papers with a strong protocol versus 34 of 66 others (protocol-only definition) — is accompanied by a power analysis so that non-significance is not misread as equivalence.

Method. Analytic two-proportion power, validated against Fisher-exact Monte Carlo simulation, is computed for the observed arm sizes and a 51.5% baseline strong-protocol rate. Under the broader definition of stringency (which also counts experimental-hardware demonstrations) the same analysis is repeated for the 13-of-22 versus 49-of-66 contrast.

Results. With 22 versus 66 papers, 80% power at $\alpha = 0.05$ requires a difference of roughly 30–35 percentage points between the families (odds ratio ≤ 0.26 or ≥ 4.0); the observed 15.2-point gap is detectable with only about 20–25% power. The broader-definition contrast is similar. The two arms' Wilson 95% intervals overlap almost completely (19.7–57.0% versus 39.7–63.2%). The honest statement is that no difference is detectable at this scale, and that meaningful differences in either direction — including a physics prior halving or doubling the odds of a strong protocol — remain compatible with these counts.

Released artifacts: power_analysis.py and power_analysis.json.

S4. Recall-audit construction, yields, and limits

The main text reports that the 1,550-record retrieval contains 16–17% of the in-scope literature cited by this review's own included reviews. This section states how that number is built.

Denominator. The deposited reference lists of the 15 included reviews and the six seed studies — 2,508 references in total — are restricted to references dated 2020 or later (1,367 are dated before 2020 and fall outside the retrieval window by design), then to references that pass the same title gate as the screening pipeline and carry matchable DOIs: 649 counted with multiplicity, 586 unique.

Numerator and result. The 1,550-record retrieval contains 112 of the 649 (17.3% per-source) and 96 of the 586 unique references (16.4%); all 96 are Tier-1 eligible. Per-query yields are design-capped, not saturated: every pass returned its top 20 records only, so the 30 query families are capped at 100 records each; the five passes yielded 3,000 fetched rows, 2,973 after document-type, year, and title-completeness filters, and 1,550 unique records after deduplication.

Limits. The denominator is a citation-conditioned sample of what the reviews happened to cite; the in-scope gate runs on titles; and references without deposited DOIs cannot be matched. The audit is therefore a floor on external validity for theme frequencies, per-family shares, and 'earliest' claims, which is why all such claims in the paper are subset- or pool-scoped (S1, rules 2 and 3). The recall miss documented in main-text Section 8.4 (the TEC 2025 GPT-4 estimator) fits this pattern exactly: it is cited by none of the included reviews either, so no reference-list audit would have surfaced it.

Released artifacts: recall_audit.py, recall_audit.json, and the cached Crossref responses under raw_cache/.

S5. Abstract-level probes

S5.1 Representativeness probe: can the pool carry the validation audit?

Question. Does the curated subset overstate validation evidence relative to the pool it was drawn from — at the one level where the 799-record pool is observable, its indexed abstracts?

Design. A seed-pinned uniform sample of 40 eligible records was drawn (released with its codes); 23 abstracts were retrievable from public indexes (Elsevier-withheld abstracts dominate the 17 missing). Each was coded under a three-value rubric: isolation or deployment protocol stated; evaluation stated without isolation; nothing stated. The pool sample's record mix: 16 primary-ML, 2 model-based, 3 review, 2 other.

Result. None of the 18 primary-looking pool records states an isolation or deployment protocol in its abstract (0 of 18), against 5 of the 38 abstract-screenable subset papers (13%); the difference is not significant (Fisher's exact $p = 0.16$) and both rates are near zero. The finding is negative in a useful way: indexed abstracts almost never carry enough protocol information to code validation strength at pool scale, in either tier, so the 48% strong-protocol share cannot be promoted from a curated-subset statistic to a field estimate without full-text access — and the scarcity of protocol statements in abstracts is itself part of what the reporting checklist of extended-review Section 9.9 (Section IX-D of the condensed manuscript) responds to.

S5.2 Baseline-comparison screen: do ML studies benchmark against model-based methods?

Question. The review had audited validation protocols but not whether ML studies compare against model-based or classical reference methods at all — a Kalman-family filter (EKF/UKF), an observer, an equivalent-circuit or physics-based model, coulomb counting, or a classical regressor such as Gaussian-process regression.

Design. A seventh, lighter classification field (added when the review's own checklist turned on its authors) records, from indexed abstracts, whether such a reference method is named as a comparator: values none, mentioned, and stated same-data comparison. It is coded from abstracts, not full texts; the released rubric (codebook.md) also defines the full-text version of the field for future work.

Result. Of the 38 abstract-screenable primary papers, 32 (84%) name no model-based or classical baseline, 2 name one without stating the comparison, and 4 (11%) state a same-data quantitative comparison — against an SVR, a 0.5C CCCV protocol, an equivalent-circuit/electrochemical model, and a rule-based controller, respectively. These are lower bounds on comparison practice (full texts may compare what abstracts omit) but exact measures of comparison visibility, and visibility is what a reader scanning the literature actually has: at abstract level, the field's answer to 'does the learned estimator beat the model-based baseline on the same data?' is, four times in five, unstated.

Released artifacts: abstract_probe.py, abstract_probe_raw.json, abstract_probe_codes.py, abstract_probe.json; rubric in codebook.md.

S6. Conference-venue probe (descriptive): does the journal-only corpus understate validation practice?

Because the corpus excludes conference proceedings — a major venue for control-oriented BMS work — the validation findings of extended-review Section 3.4 (condensed Section III-C) could in principle be an artifact of venue selection: if conference papers systematically demonstrate more hardware and hardware-in-the-loop validation than journals, the journal-only sample would understate the field's

actual practice. To probe this, we curated a sensitivity set of 14 battery papers from six control and transportation-electrification venues (ACC, ECC, CCDC, APEC, VPPC, ITEC), retrieved by venue-scoped Crossref queries, restricted to proceedings articles, ranked by citation count, and verified by the same DOI-resolution procedure as the main corpus (released in full as conference_sensitivity.csv). Two scope boundaries are stated before the counts. First, the venue set does not include CDC or IFAC proceedings, themselves major control venues, so the check probes part of the conference literature rather than censusing it. Second, the set's retrieval window (2019–2025) starts one year earlier than the main pool's 2020–2026 window, because the venue-scoped queries carried no lower year bound: three of the 14 papers are dated 2019, and excluding them leaves 11 papers of which 6 are learning-based, with unchanged hardware (2), public-dataset (2), and simulation (1) counts among the learning-based papers — only the not-stated class shrinks, from three papers to one. The comparison is descriptive and abstract-level throughout — indexed abstracts versus the full-text coding of the main corpus — so it is reported as a labelled descriptive check rather than a validated sensitivity analysis, and none of its counts enter the headline statistics of the main text. A pre-stated inclusion rule was then applied mechanically to all retrieved proceedings papers (the table below lists all 14 with the rule's outcome): a paper enters the learning-path comparison only if machine learning is a substantive component of its contribution. Under this rule 8 of the 14 qualify; the remainder are conventional model-based estimators (Kalman-filter variants, multiple-model adaptive estimation, batch least squares), an EKF-critique, and one overview, and all 14 are released so that the rule can be audited rather than trusted. Validation practice was classified conservatively from the indexed abstracts into four classes: hardware/experimental stated, public dataset, simulation-level, and not stated.

Table S1 Conference-path descriptive set: all 14 battery papers retrieved from control and transportation-electrification venues (2019–2025), Crossref-verified, with validation evidence stated in their indexed abstracts. The pre-stated inclusion rule (machine learning as a substantive component of the contribution) retains 8 of the 14 for the learning-path comparison; the remainder are conventional model-based estimators or an overview and are retained here for transparency.

Venue (year)	Function	Method (short)	Learning-based?	Validation evidence stated in indexed abstract
ITEC 2019	SOC	LSTM recurrent network with transfer learning doi: 10.1109/itec.2019.8790543	Yes	Not stated
VPPC 2019	SOC	SVR + PCA with dual-polarization model doi: 10.1109/vppc46532.2019.8952458	Yes	Not stated
VPPC 2019	SOC	Kalman filter on reduced electrochemical model doi: 10.1109/vppc46532.2019.8952284	No (model-based)	Simulation-level
ACC 2021	SOC	Multiple-model adaptive estimation doi: 10.23919/acc50511.2021.9482734	No (model-based)	Simulation-level
VPPC 2021	SOC	Computationally efficient neural network doi: 10.1109/vppc53923.2021.9699201	Yes	Simulation-level
VPPC 2021	SOH	ML ensemble doi: 10.1109/vppc53923.2021.9699273	Yes	Public dataset
ECC 2021	SOH	Reinforcement learning + observer doi: 10.23919/ecc54610.2021.9655199	Yes	Not stated
CCDC 2022	SOH	CNN-BiLSTM transfer learning doi: 10.1109/ccdc55256.2022.10033695	Yes	Hardware (stated); public data
VPPC 2022	Prognostics	ML on aging data doi: 10.1109/vppc55846.2022.10003444	Yes	Hardware (stated)
ITEC 2022	SOC+SOH	Dual extended Kalman filter doi: 10.1109/itec53557.2022.9813961	No (model-based)	Not stated
ACC 2023	SOC+SOH	Real-time estimator + batch least	No (model-	Not stated

		squares doi: 10.23919/acc55779.2023.10156326	based)	
APEC 2023	Overview	Review of ML-enabled estimation doi: 10.1109/apec43580.2023.10131605	Overview	Not applicable
ITEC 2023	SOH	Histogram data + PCA + regression doi: 10.1109/itec55900.2023.10187012	Yes	Public dataset
ACC 2024	SOC+SOH	EKF critique and improvement doi: 10.23919/acc60939.2024.10644628	No (model-based)	Simulation-level

The comparison finds no evidence of a venue gap in stated validation evidence, and with eight abstracts on one side it cannot rule one out. Of the 8 learning-based conference papers, 2 state hardware or experimental validation in their indexed abstracts, 2 rest on public datasets, 1 describes simulation-level validation, and 3 state no validation basis at all, alongside 20 of the 88 primary-application journal papers with experimental-hardware demonstrations; at this scale the difference in stated-evidence rates between the two paths is not interpretable, so no percentages are read off it. Two caveats bound whatever reading is put on these counts: conference abstracts are frequently truncated in bibliographic indexes, so the comparison measures stated evidence rather than underlying practice; and the sensitivity set is a curated sample, not a census. The comparison therefore remains descriptive — eight truncated abstracts cannot carry an inference about venue populations — so the venue question stays open on this evidence, and the reporting standards proposed in extended-review Section 9.9 (condensed Section IX-D) apply to journal and conference venues alike.

Released artifacts: conference_sensitivity.csv, conference_verified.json, and the venue-scoped retrieval scripts; cited proceedings papers are listed by DOI in Table S1.

S7. Validation audit: per-paper evidence for the challenged and upgraded labels

This section carries the per-paper mechanics behind the audit summary of extended-review Sections 2.5 and 3.4 (condensed Sections II-C and III-C), so that the main text states results only. Three layers are involved: the round-1 blind re-coding (abstract-level agreement), the round-2 re-coding under codebook v1.1 (rule-application agreement on a shared evidence base), and the post-freeze evidence-side re-code by the AI research assistant declared in the AI-use statement (coder D), which audits the frozen labels from the evidence side rather than measuring coder agreement.

S7.1 Coder D re-code: the four challenged strong labels and two upgrades

After the label set was frozen, coder D attempted the validation field of all 88 primary papers, covering 46 with retrievable evidence (7 full texts, 39 abstracts) and determinably coding 23, from locked evidence sheets and before consulting the released per-paper labels, applying the codebook's absorbing rule and a dedicated 'not reported' code. Of the 14 determinable strong labels, 10 are confirmed and 4 are contradicted: (i) the graph-network SOH study (doi 10.1016/j.est.2024.113502) is verified, from its accepted manuscript, to train on the first 70% of each cell's cycles and reserve the remaining 30% of the same cells for testing — a within-cell split, coded cross-cell in the released labels; (ii) the ECM-to-electrochemical-parameter study (doi 10.1016/j.est.2024.113257) states validation on a synthetic dataset generated by a pseudo-two-dimensional model (coded cross-cell in the released labels); and (iii–iv) two further strong labels (doi 10.1016/j.egyai.2022.100188 and doi 10.1016/j.est.2026.123023) could not be confirmed from abstract-level evidence, their full texts being paywalled. Two determinable non-strong labels would be upgraded: the varying-temperature SOC

study (doi 10.1016/j.energy.2020.117664), whose abstract states estimation at temperatures unseen in training (the codebook's cross-domain clause), and the physics-informed RL study (doi 10.1038/s41598-025-30602-4), which reports a NASA-to-CALCE transfer as a robustness check (an absorbing-rule judgment call). Bucket agreement on the determinable subset is 17 of 23 (74%, κ = 0.47, Wilson 95% CI 53.5–87.5%). No released label was altered: the four challenges and two upgrades are itemized with the evidence that motivated them, the 42-of-88 count remains the lead author's single full-text coding with its strongest challenges on record rather than latent, and the two label corrections the itemization supports are left as flagged author decisions rather than silent edits.

S7.2 Round-2 full-text re-coding: five candidate label corrections

The round-2 full-text re-coding reaches the same synthetic-data reading of the ECM-to-electrochemical-parameter study (doi 10.1016/j.est.2024.113257) by an independent route and itemizes four further candidate label corrections — two Severson-dataset papers whose released chemistry records NMC where the dataset itself is LFP, one LCO chemistry record, and one study whose released labels disagree with its own released code repository — again left unapplied, with the evidence released. Together with S7.1 these constitute the complete discrepancy record against the frozen labels.

Released artifacts: corpus/validation_recode/ (validation_recode.json with the per-paper coder-D evidence), coder_B_30_fulltext.csv, coder_C_30_fulltext.csv, dispute_items_fulltext.csv, agreement_report_fulltext.csv, and the hash-tracked reliability log under corpus/reliability/.

All scripts run from the repository root against the released data and reproduce their outputs deterministically (seeded where stochastic). Deposit of record: Zenodo, DOI 10.5281/zenodo.23436202; development mirror: the GitHub repository of the same name.